15-322 / 15-622 | FALL 2026

# INTRODUCTION TO COMPUTER MUSIC

NOW FULLY IN PYTHON

## TURN CODE INTO SOUND

TUE / THU
11:00 AM—12:20 PM

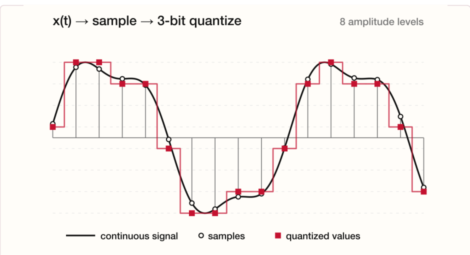
CIC 1203

9 UNITS (15-322)
12 UNITS (15-622)

## WHAT YOU'LL LEARN

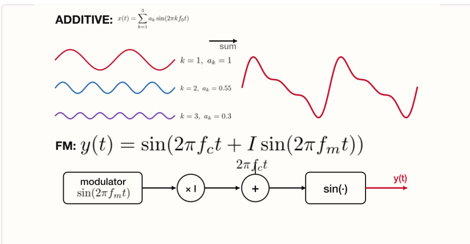
### DIGITAL AUDIO
01

SAMPLING · QUANTIZATION · ALIASING



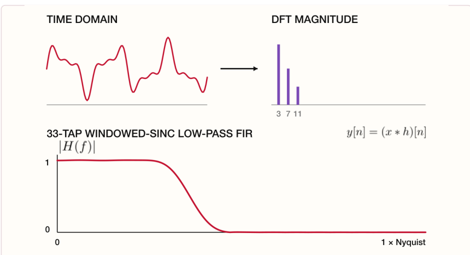

### SYNTHESIS
02

ADDITIVE · FM · SAMPLE-BASED · PHYSICAL MODELING



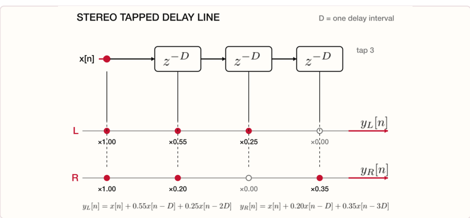

### FOURIER & DSP
03

DFT / FFT · FILTERS · CONVOLUTION · STFT



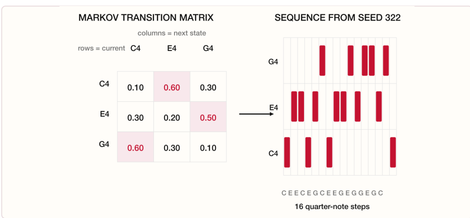

### EFFECTS & SPATIALIZATION
04

DELAY LINES · MODULATION · PHASE VOCODING



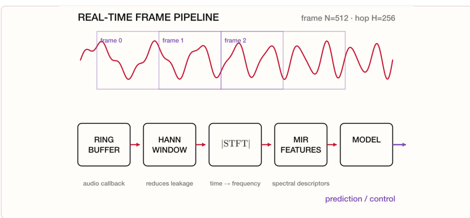

### ALGORITHMIC COMPOSITION
05

MARKOV CHAINS · LPC · MIDI




### REAL-TIME AUDIO & MUSIC AI
06

STREAMING · MIR · MODELS




9 WEEKLY ASSIGNMENTS
+ FINAL PROJECT

PREREQUISITE:
15-122 OR 15-112

MUSIC EXPERIENCE
NOT REQUIRED

INSTRUCTOR: CHRIS DONAHUE

CARNEGIE MELLON UNIVERSITY